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§ = Authors ordered by contribution

- [74] * Gregorio de la Fuente and Jesse Thaler, *Simplex Demixing: Disentangling Multiple Light-Flavor Jets at Colliders* [arXiv:2607.24921].
- [73] Benjamin Nachman and Jesse Thaler, *The Well-Tempered Likelihood: Honest Confidence Intervals for Misspecified Models* [arXiv:2607.20718].
- [72] § Rikab Gambhir, Luisa Lucie-Smith, and Jesse Thaler, *Interpreting 'Interpretability' and Explaining 'Explainability' in Machine Learning in Physics* [arXiv:2606.26228].
- [71] § Umar Sohail Qureshi, Krish Desai, Jesse Thaler, and Benjamin Nachman, *Reweighting Adversarial Networks for Unbinned Unfolding* [arXiv:2606.06603].
- [70] Ezequiel Alvarez, Sean Benevedes, Manuel Szewc, and Jesse Thaler, *Many Wrongs Make a Right: Leveraging Biased Simulations Towards Unbiased Parameter Inference* [arXiv:2604.02219].
- [69] Samuel Bright-Thonney, Thomas R. Harvey, Andre Lukas, and Jesse Thaler, *Sven: Singular Value Descent as a Computationally Efficient Natural Gradient Method* [arXiv:2604.01279].
- [68] Nathan Benjamin, A. Liam Fitzpatrick, Wei Li, and Jesse Thaler, *Descending into the Modular Bootstrap* [arXiv:2604.01275].
- [67] Benoît Assi, Kyle Lee, and Jesse Thaler, *Improving Parton Shower Predictions via Precision Moments of Energy Flow Polynomials* [arXiv:2604.00084].
- [66] Andrew Ferguson, Marisa LaFleur, Lars Ruthotto, Jesse Thaler, Yuan-Sen Ting, Pratyush Tiwary, Soledad Villar, et al., *The Future of Artificial Intelligence and the Mathematical and Physical Sciences (AI+MPS)*, MLST 7:023001 (2026) [arXiv:2509.02661].
- [65] * Sean Benevedes and Jesse Thaler, *Frequentist Uncertainties on Neural Density Ratios with WiFi Ensembles*, Phys. Rev. D112:056024 (2025) [arXiv:2506.00113].
- [64] * Rikab Gambhir, Radha Mastandrea, Benjamin Nachman, Jesse Thaler, *Isolating Unisolated Upsilons with Anomaly Detection in CMS Open Data*, Phys. Rev. Lett. 135:021902 (2025) [arXiv:2502.14036].
- [63] Benoît Assi, Stefan Höche, Kyle Lee, and Jesse Thaler, *QCD Theory Meets Information Theory*, Phys. Rev. Lett. 135:131901 (2025) [arXiv:2501.17219].

- [62] Johann Brehmer, Víctor Bresó, Pim de Haan, Tilman Plehn, Huilin Qu, Jonas Spinner, and Jesse Thaler, *A Lorentz-Equivariant Transformer for All of the LHC*, SciPost Physics 19:108 (2025) [arXiv:2411.00446].
- [61] Jesse Thaler and Sokratis Trifinopoulos, *Flavor Patterns of Fundamental Particles from Quantum Entanglement?*, Phys. Rev. D111:056021 (2025) [arXiv:2410.23343].
- [60] * Samuel Alipour-fard, Ankita Budhraj, Jesse Thaler, and Wouter J. Waalewijn, *New Angles on Energy Correlators*, Phys. Rev. Lett. 134:231902 (2025) [arXiv:2410.16368].
- [59] * Rikab Gambhir, Andrew J. Larkoski, and Jesse Thaler, *SPECTER: Efficient Evaluation of the Spectral EMD*, JHEP 2412:219 (2024) [arXiv:2410.05379].
- [58] Krish Desai, Benjamin Nachman, and Jesse Thaler, *Moment Unfolding*, Phys. Rev. D110:116013 (2024) [arXiv:2407.11284].
- [57] § Jonas Spinner, Victor Bresó, Pim de Haan, Tilman Plehn, Jesse Thaler, and Johann Brehmer, *Lorentz-Equivariant Geometric Algebra Transformers for High-Energy Physics*, NeurIPS 2024 [arXiv:2405.14806].
- [56] * Rikab Gambhir, Athis Osathapan, and Jesse Thaler, *Moments of Clarity: Streamlining Latent Spaces in Machine Learning using Moment Pooling*, Phys. Rev. D110:074020 (2024) [arXiv:2403.08854].
- [55] § Siddharth Mishra-Sharma, Yiding Song, and Jesse Thaler, *PAPERCLIP: Associating Astronomical Observations and Natural Language with Multi-Modal Models*, Conf. Lang. Mod. 2024 [arXiv:2403.08851].
- [54] † Eric M. Metodiev, Jesse Thaler, and Raymond Wynne, *Anomaly Detection in Collider Physics via Factorized Observables*, Phys. Rev. D110:055012 (2024) [arXiv:2312.00119].
- [53] Samuel Bright-Thonney, Benjamin Nachman, and Jesse Thaler, *Safe but Incalculable: Energy-Weighting is Not All You Need*, Phys. Rev. D110:014029 (2024) [arXiv:2311.07652].
- [52] Fabrizio Caola, Radosław Grabarczyk, Maxwell L. Hutt, Gavin P. Salam, Ludovic Scyboz, and Jesse Thaler, *Flavoured Jets with Exact Anti-kt Kinematics and Tests of Infrared and Collinear Safety*, Phys. Rev. D108:094010 (2023) [arXiv:2306.07314].
- [51] Andrew J. Larkoski and Jesse Thaler, *A Spectral Metric for Collider Geometry*, JHEP 2308:107 (2023) [arXiv:2305.03751].
- [50] * Samuel Alipour-fard, Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Pileup and Infrared Radiation Annihilation (PIRANHA): A Paradigm for Continuous Jet Grooming*, JHEP 2309:157 (2023) [arXiv:2305.00989].
- [49] * Demba Ba, Akshunna S. Dogra, Rikab Gambhir, Abiy Tasissa, and Jesse Thaler, *SHAPER: Can You Hear the Shape of a Jet?*, JHEP 2306:195 (2023) [arXiv:2302.12266].
- [48] Erik Buhmann, Gregor Kasieczka, and Jesse Thaler, *EPiC-GAN: Equivariant Point Cloud Generation for Particle Jets*, SciPost Phys. 15:130 (2023) [arXiv:2301.08128].
- [47] Peter Onyisi, Delon Shen, and Jesse Thaler, *Comparing Point Cloud Strategies for Collider Event Classification*, Phys. Rev. D108:012001 (2023) [arXiv:2212.10659].

- [46] [†] Eric R. Anschuetz, Lena Funcke, Patrick T. Komiske, Serhii Kryhin, and Jesse Thaler, *Degeneracy Engineering for Classical and Quantum Annealing: A Case Study of Sparse Linear Regression in Collider Physics*, Phys. Rev. D106:056008 (2022) [arXiv:2205.10375].
- [45] Pedro Cal, Jesse Thaler, and Wouter J. Waalewijn, *Power Counting Energy Flow Polynomials*, JHEP 2209:021 (2022) [arXiv:2205.06818].
- [44] * Rikab Gambhir, Benjamin Nachman, and Jesse Thaler, *Bias and Priors in Machine Learning Calibrations for High Energy Physics*, Phys. Rev. D106:036011 (2022) [arXiv:2205.05084].
- [43] [†] Patrick T. Komiske, Serhii Kryhin, and Jesse Thaler, *Disentangling Quarks and Gluons with CMS Open Data*, Phys. Rev. D106:094021 (2022) [arXiv:2205.04459].
- [42] * Rikab Gambhir, Benjamin Nachman, and Jesse Thaler, *Learning Uncertainties the Frequentist Way: Calibration and Correlation in High Energy Physics*, Phys. Rev. Lett. 129:082001 (2022) [arXiv:2205.03413].
- [41] Hao Chen, Ian Moulton, Jesse Thaler, and Hua Xing Zhu, *Non-Gaussianities in Collider Energy Flux*, JHEP 2207:146 (2022) [arXiv:2205.02857].
- [40] Andrea Delgado and Jesse Thaler, *Quantum Annealing for Jet Clustering with Thrust*, Phys. Rev. D106:094016 (2022) [arXiv:2205.02814].
- [39] Patrick T. Komiske, Ian Moulton, Jesse Thaler, and Hua Xing Zhu, *Analyzing N-point Energy Correlators Inside Jets with CMS Open Data*, Phys. Rev. Lett. 130:051901 (2023) [arXiv:2201.07800].
- [38] Krish Desai, Benjamin Nachman, and Jesse Thaler, *SymmetryGAN: Symmetry Discovery with Deep Learning*, Phys. Rev. D105:096031 (2022) [arXiv:2112.05722].
- [37] Benjamin Nachman and Jesse Thaler, *Neural Conditional Reweighting*, Phys. Rev. D105:076015 (2022) [arXiv:2107.08979].
- [36] Benjamin Nachman and Jesse Thaler, *E Pluribus Unum Ex Machina: Learning from Many Collider Events at Once*, Phys. Rev. D103:116013 (2021) [arXiv:2101.07263].
- [35] Taylor Faucett, Jesse Thaler, and Daniel Whiteson, *Mapping Machine-Learned Physics into a Human-Readable Space*, Phys. Rev. D103:036020 (2021) [arXiv:2010.11998].
- [34] Jasmine Brewer, Jesse Thaler, and Andrew P. Turner, *Data-Driven Quark and Gluon Jet Modification in Heavy-Ion Collisions*, Phys. Rev. C103:L021901 (2021) [arXiv:2008.08596].
- [33] Benjamin Nachman and Jesse Thaler, *Neural Resampler for Monte Carlo Reweighting with Preserved Uncertainties*, Phys. Rev. D102:076004 (2020) [arXiv:2007.11586].
- [32] Cari Cesarotti and Jesse Thaler, *A Robust Measure of Event Isotropy at Colliders*, JHEP 2008:084 (2020) [arXiv:2004.06125].
- [31] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *The Hidden Geometry of Particle Collisions*, JHEP 2007:006 (2020) [arXiv:2004.04159].
- [30] * Anders Andreassen, Patrick T. Komiske, Eric M. Metodiev, Benjamin Nachman, and Jesse Thaler, *OmniFold: A Method to Simultaneously Unfold All Observables*, Phys. Rev. Lett. 124:182001 (2020) [arXiv:1911.09107].

- [29] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Cutting Multiparticle Correlators Down to Size*, Phys. Rev. D101:036019 (2020) [arXiv:1911.04491].
- [28] Timothy Cohen, Gilly Elor, Andrew J. Larkoski, and Jesse Thaler, *Circumnavigating Collinear Superspace*, JHEP 2002:156 (2020) [arXiv:1909.00009].
- [27] § Annie Y. Wei, Preksha Naik, Aram W. Harrow, and Jesse Thaler, *Quantum Algorithms for Jet Clustering*, Phys. Rev. D101:094015 (2020) [arXiv:1908.08949].
- [26] *† Patrick T. Komiske, Radha Mastandrea, Eric M. Metodiev, Preksha Naik, and Jesse Thaler, *Exploring the Space of Jets with CMS Open Data*, Phys. Rev. D101:034009 (2020) [arXiv:1908.08542].
- [25] Cari Cesarotti, Yotam Soreq, Matthew J. Strassler, Jesse Thaler, and Wei Xue, *Searching in CMS Open Data for Dimuon Resonances with Substantial Transverse Momentum*, Phys. Rev. D100:015021 (2019) [arXiv:1902.04222].
- [24] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *The Metric Space of Collider Events*, Phys. Rev. Lett. 123:041801 (2019) [arXiv:1902.02346].
- [23] Jasmine Brewer, José Guilherme Milhano, and Jesse Thaler, *Sorting Out Quenched Jets*, Phys. Rev. Lett. 122:222301 (2019) [arXiv:1812.05111].
- [22] Timothy Cohen, Gilly Elor, Andrew J. Larkoski, and Jesse Thaler, *Navigating Collinear Superspace*, JHEP 2002:146 (2020) [arXiv:1810.11032].
- [21] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Energy Flow Networks: Deep Sets for Particle Jets*, JHEP 1901:121 (2019) [arXiv:1810.05165].
- [20] § Hongwan Liu, Brodi D. Elwood, Matthew Evans, and Jesse Thaler, *Searching for Axion Dark Matter with Birefringent Cavities*, Phys. Rev. D100:023548 (2019) [arXiv:1809.01656].
- [19] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *An Operational Definition of Quark and Gluon Jets*, JHEP 1811:059 (2018) [arXiv:1809.01140].
- [18] † Eleanor Hall and Jesse Thaler, *Photon Isolation and Jet Substructure*, JHEP 1809:164 (2018) [arXiv:1805.11622].
- [17] * Benjamin T. Elder and Jesse Thaler, *Aspects of Track-Assisted Mass*, JHEP 1903:104 (2019) [arXiv:1805.11109].
- [16] Frédéric A. Dreyer, Lina Necib, Gregory Soyez, and Jesse Thaler, *Recursive Soft Drop*, JHEP 1806:093 (2018) [arXiv:1804.03657].
- [15] * Eric M. Metodiev and Jesse Thaler, *On the Topic of Jets: Disentangling Quarks and Gluons at Colliders*, Phys. Rev. Lett. 120:241602 (2018) [arXiv:1802.00008].
- [14] * Patrick T. Komiske, Eric M. Metodiev, and Jesse Thaler, *Energy Flow Polynomials: A Complete Linear Basis for Jet Substructure*, JHEP 1804:013 (2018) [arXiv:1712.07124].
- [13] Evan Coleman, Marat Freytsis, Andreas Hinzmann, Meenakshi Narain, Jesse Thaler, Nhan Tran, and Caterina Vernieri, *The Importance of Calorimetry for Highly-Boosted Jet Substructure*, JINST 13:T01003 (2018) [arXiv:1709.08705].
- [12] * Eric M. Metodiev, Benjamin Nachman, and Jesse Thaler, *Classification Without Labels: Learning from Mixed Samples in High Energy Physics*, JHEP 1710:174 (2017) [arXiv:1708.02949].

- [11] [†] Christopher Frye, Andrew J. Larkoski, Jesse Thaler, and Kevin Zhou, *Casimir Meets Poisson: Improved Quark/Gluon Discrimination with Counting Observables*, JHEP 1709:085 (2017) [arXiv:1704.06266].
- [10] ^{†§} Aashish Tripathy, Wei Xue, Andrew Larkoski, Simone Marzani, and Jesse Thaler, *Jet Substructure Studies with CMS Open Data*, Phys. Rev. D96:074003 (2017) [arXiv:1704.05842].
- [9] ^{*†} Benjamin T. Elder, Massimiliano Procura, Jesse Thaler, Wouter J. Waalewijn, and Kevin Zhou, *Generalized Fragmentation Functions for Fractal Jet Observables*, JHEP 1706:085 (2017) [arXiv:1704.05456].
- [8] [†] Andrew Larkoski, Simone Marzani, Jesse Thaler, Aashish Tripathy, and Wei Xue, *Exposing the QCD Splitting Function with CMS Open Data*, Phys. Rev. Lett. 119:132003 (2017) [arXiv:1704.05066].
- [7] Philippe Gras, Stefan Höche, Deepak Kar, Andrew Larkoski, Leif Lönnblad, Simon Plätzer, Andrzej Siódmok, Peter Skands, Gregory Soyez, and Jesse Thaler, *Systematics of Quark/Gluon Tagging*, JHEP 1707:091 (2017) [arXiv:1704.03878].
- [6] Yevgeny Kats, Matthew McCullough, Gilad Perez, Yotam Soreq, and Jesse Thaler, *Colorful Twisted Top Partners and Partnerium at the LHC*, JHEP 1706:126 (2017) [arXiv:1704.03393].
- [5] Philip Ilten, Nicholas L. Rodd, Jesse Thaler, and Mike Williams, *Disentangling Heavy Flavor at Colliders*, Phys. Rev. D96:054019 (2017) [arXiv:1702.02947].
- [4] ^{*} Ian Moulton, Lina Necib, and Jesse Thaler, *New Angles on Energy Correlation Functions*, JHEP 1612:153 (2016) [arXiv:1609.07483].
- [3] Fabio Maltoni, Michele Selvaggi, and Jesse Thaler, *Resurrecting the Dead Cone*, Phys. Rev. D94:054015 (2016) [arXiv:1606.03449].
- [2] Philip Ilten, Yotam Soreq, Jesse Thaler, Mike Williams, and Wei Xue, *Inclusive Dark Photon Search at LHCb*, Phys. Rev. Lett. 116:251803 (2016) [arXiv:1603.08926].
- [1] Yonatan Kahn, Benjamin R. Safdi, and Jesse Thaler, *Broadband and Resonant Approaches to Axion Dark Matter Detection*, Phys. Rev. Lett. 117:141801 (2016) [arXiv:1602.01086].